# 4008. Minimum Initial Strength to Defeat All Monsters

Solved

Medium Hint

You are given an integer array `monsters`, where `monsters[i]` represents the strength of the $i^{th}$ monster.

You are also given a 2D integer array `boosts`, where `boosts[i] = [l_i, r_i, v_i]` indicates that $v_i$ is added to your **temporary bonus** while fighting any monster whose index lies in $[l_i, r_i]$. Boost ranges may overlap, and the values of all applicable boosts are added together.

You start with a **non-negative** initial strength and fight the monsters from left to right.

For each monster at index `i`:

- Let `bonus` be the **sum** of the values of all boosts that apply to monster `i`.
- You can defeat the monster only if your current strength plus `bonus` is **at least** `monsters[i]`.
- After defeating the monster, only your current strength decreases by `monsters[i]`. If it becomes **negative**, it is set to 0.

Return the **minimum** initial strength required to defeat all monsters.

Note: The temporary bonus is used only to determine whether the current monster can be defeated. It does not otherwise change your current strength.

**Example 1:**

> **Input:** monsters = [5,10,15], boosts = [[1,1,10]]
>
> **Output:** 30
>
> **Explanation:**
>
> Let's start with an initial strength of 30.
>
> - `monsters[0] = 5`: At index 0, the bonus is 0. Since `30 + 0 >= 5`, this monster can be defeated. The strength becomes `30 - 5 = 25`.
> - `monsters[1] = 10`: At index 1, the bonus is 10. Since `25 + 10 >= 10`, this monster can be defeated. The strength becomes `25 - 10 = 15`.
> - `monsters[2] = 15`: At index 2, the bonus is 0. Since `15 + 0 >= 15`, this monster can be defeated. The strength becomes `15 - 15 = 0`.
>
> Thus, the minimum initial strength required is 30.

**Example 2:**

> **Input:** monsters = [5,10,15], boosts = [[1,2,10],[1,2,5]]
>
> **Output:** 5
>
> **Explanation:**
>
> Let's start with an initial strength of 5.
>
> - `monsters[0] = 5`: The bonus is 0. Since `5 + 0 >= 5`, the monster can be defeated. The strength becomes `5 - 5 = 0`.
> - `monsters[1] = 10`: The two overlapping boosts provide `bonus = 10 + 5 = 15`. Since `0 + 15 >= 10`, the monster can be defeated. The strength remains 0.
> - `monsters[2] = 15`: The two overlapping boosts again provide `bonus = 15`. Since `0 + 15 >= 15`, the monster can be defeated. The strength remains 0.
>
> Thus, the minimum initial strength required is 5.

**Constraints:**

- $1 <= monsters.length <= 5 * 10^4$
- $1 <= monsters[i] <= 10^9$
- $0 <= boosts.length <= 5 * 10^4$
- $boosts[i] == [l_i, r_i, v_i]$
- $0 <= l_i <= r_i < monsters.length$
- $1 <= v_i <= 10^9$

Seen this question in a real interview before? 1/6

Yes No

Accepted 16,821/33.5K | Acceptance Rate 50.2%

Topics

Hint 1

Hint 2

Hint 3

Discussion (36)